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Chemical constituents of selected unifloral Greek bee-honeys with antimicrobial activity

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ABSTRACT

(±)-3-Hydroxy-4-phenyl-2-butanone (**1**) and (+)-8-hydroxylinalool (**2**) were isolated from Greek honey samples and their structures were determined by spectroscopic and mass spectrometry methods. The enantiomeric ratio of **1** and **2** was determined using chiral GC–MS. Compound **1** has recently been proposed as a possible chemical marker of thyme honey, but although **1**, in our hands, was identified only in monofloral thyme honey amongst the honey samples, it was not found in all studied thyme honey samples and additionally its origin was not directly associated with thyme, proving that compound **1** should not be considered as an unambiguous marker of thyme honey. Compound **2** was isolated and identified as the (+)-6S isomer only in monofloral citrus honey. The combination of (+)-8-hydroxylinalool with methyl anthranilate and caffeine could be proposed as a fingerprint marker for the description of citrus honey. All studied honey extracts and the isolated compounds were also tested for their antimicrobial activity.

1. Introduction

Bee-honey has been a famous food through antiquity and its health benefits are of increasing interest. Several classical texts of ancient Greece, e.g. Homer's Iliad and Odyssey, the Deipnosophists of Athenaeus, as well as philosophical texts of Plato, appreciate the benefits of honey consumption for human's health.

The Greek flora present a generally known biodiversity with a high percentage of endemic plants. Honeybees forage on plants over a relatively large area, each of them visiting thousands of flowers per day and therefore honey is most often a mixture of different kinds of nectars and/or honeydews. Unifloral honeys are not as common as multifloral. The production of the first group presents technical difficulties, and depends on the beekeepers know-how and also on the early blooming or honeydewing period of certain flowers or trees. Usually, honey is considered as unifloral when the pollen frequency of one plant is over 45% (Tomas-Barberan, Martos, Ferreres, Radovic, & Anklam, 2001). Many unifloral honey types are known around the world whilst, in Greece, thyme, orange blossom (*Citrus* spp.), fir and pine are amongst the most common unifloral honey types. Fir and pine honeys are categorised as honeydew honeys and citrus and thyme as nectar honeys (floral honeys). Floral honey is derived from the nectar of flowering plants whilst honeydew honey is obtained from honeydew secreted by the living parts of plants or excreted onto them by sap-sucking insects, e.g. *Marshalina hellenica* and *Physokermes hemicryphus* for pine and fir honeys, respectively.

The floral origin of honey significantly affects the organoleptic character and the biological and economic value of honey and, for this reason, it is very important to find reliable techniques for the verification of the floral source of each honey type.

The floral origin of honey can be found by pollen analysis (melissopalinology) but this technique is complicated and very dependent on the ability of the expert. Moreover, in some cases, e.g. citrus honey, pollen analysis cannot be used for floral verification because the amount of the pollen in the honey is much lower than that of the corresponding nectar contribution. In addition the determination of the botanical origin is more difficult in honeydew honeys, because determination of such honey-types cannot be carried out by palynological analyses. In these cases, the best identification method of their floral source could be the use of chemical markers easily identified in their organic extracts by analytical methods, e.g. GC–MS and/or HPLC–MS.

Several previous works have satisfactorily correlated the presence of specific chemical constituents with the corresponding floral source. Some of these studies have been based on non-volatile secondary metabolites (e.g. hesperetin for Citrus honey (Ferreres, Garcia-Viguera, Tomas-Lorente, & Tomas-Barberan, 1993)), whilst other studies use only volatile constituents. A number of authors have proposed that specific volatile compounds are characteristic of a given floral origin, and may be considered as distinguishing "floral markers", e.g. 3-methylbutanal, 3-hydroxy-4,5-dimethyl-2(5H)-furanone (sotolon) and (E)-β-damascenone for buckwheat honey

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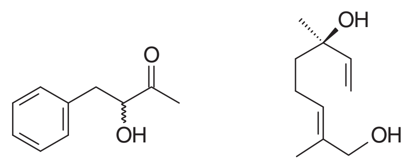

Fig. 1. The structures of (±)-3-hydroxy-4-phenyl-2-butanone (**1**) and (+)-8-hydroxylinalool (**2**).

(Zhou, Wintersteen, & Cadwallader, 2002), 2-hydroxyacetophenone and eucarvone for Chestnut honey (Castro-Vazquez, Diaz-Maroto, de Torres, & Perez-Coello, 2010) and methyl salicylate for willow honeydew honey (Jercovic, Marijanovic, Tuberoso, Bubalo, & Kezic, 2010).

In the framework of our scientific studies on Greek bee-keeping products (Melliou & Chinou, 2005; Melliou, Stratis, & Chinou, 2007) and, in an effort to identify volatile constituents restricted in specific Greek honey types, we performed a chemical screening of fifteen unifloral Greek honey samples (thyme, pine, citrus, gossypium, fir), from thirteen regions of Greece. We report, in this study, the isolation, structure elucidation, enantiomeric ratio and antimicrobial activities of (±)-3-hydroxy-4-phenyl-2-butanone (**1**) and (+)-8-hydroxylinalool (**2**) (Fig. 1) which had previously been reported as characteristic constituents restricted in some of the studied bee-honeys. For comparative reasons, some honey samples from Poland, from completely different floral sources, were also screened.

2. Materials and methods

2.1. General experimental procedures

Optical rotations were measured with a Perkin–Elmer 341 polarimeter. NMR spectra were recorded on Bruker DRX 400 and Bruker AC 200 spectrometers [^{1}H (400 and 200 MHz) and ^{13}C (50 MHz)]; chemical shifts are expressed in ppm downfield from TMS.

GC–MS analysis was carried out on a Hewlett–Packard 6890-5973 system operating in EI mode, equipped with a 30 m × 0.25 mm i.d.; 0.25 μm HP-5 MS capillary column; temperature programme: 60 °C (5 min) to 280 °C at a rate of 3 °C/min; injection temperature 200 °C.

Chiral GC–MS was carried out on a Finnigan GCQ *Plus* mass spectrometer operating in EI mode equipped with a 30 m × 0.25 mm i.d.; 0.25 μm b-DEX se (RESTEK) capillary column; temperature programme: 60 °C to 220 °C at a rate of 3 °C/min; injection temperature 200 °C.

Medium pressure liquid chromatography (MPLC) was performed with a Büchi model 688 apparatus on columns containing Si gel 60 Merck (20–40 μm). Thin-layer chromatography (TLC) was performed on plates coated with Si gel 60 F_{254} Merck, 0.25 mm.

2.2. Sample collection

The samples of Greek honeys were provided by certified Greek bee-keepers, in 2003, and characterised by pollen analysis, through microscopy, by Dr. Karabournioti. The honey floral sources and regions of collection are given in Table 1. Samples A and K were used for the isolation of the pure compounds. The whole procedure, of the isolation of the secondary metabolites, was repeated after recollection of honey samples during the years 2003, 2006 and 2009.

Orange (*Citrus sinensis* CV, Washington) flowers were collected in May, 2004, from Arboriculture Station at Poros Island, Greece, and identified by the Director of the Station Mrs. Th. Agorastou.

Table 1
Studied monofloral honey samples.

Sample	Honey floral source	Origin
A	Thyme	Kythira
B	Thyme	Kea
C	Thyme	Rhodes
D	Thyme	Mani
E	Thyme	Kos
F	Thyme	Crete
G	Pine	Rhodes
H	Pine	Evia
I	Pine	Macedonia
J	Citrus	Argos
K	Citrus	Lakonia
L	Citrus	Arta
M	Gossypium	Macedonia
N	Abies	Vitina
O	Abies	Karpenisi

Spartium junceum and *Coridothymus capitatus* flowers were collected in May 2006 and 2009, in the Attiki region, Greece, and identified by Dr. E. Kalpoutzakis. Samples have been deposited at the Herbarium of the Laboratory of Pharmacognosy and Chemistry of Natural Products.

Unifloral Polish honey samples were collected from Ostrowsko in 2004 and were kindly provided by Prof. K. Glowniak.

2.3. Extraction and isolation of compounds

2.3.1. Honey extraction

A quantity (100 g) of each honey sample was diluted with water (80 ml) and extracted first with petroleum ether (150 ml) and then with EtOAc (150 ml). Each organic layer was separately collected and evaporated.

2.3.2. Flower extraction

A quantity (100 g) of fresh flowers of each species was extracted with 150 ml of petroleum ether (for *Spartium*, and thyme flowers) or dichloromethane (for *Citrus* flowers) under ultrasound sonication for 30 min. The solvent was filtered, dried over Na_2SO_4 and gently evaporated to 2 ml. The final solution was placed in a vial, sealed and kept at 4 °C until used for GC–MS analysis.

2.3.3. Isolation

(+)-8-Hydroxylinalool (**2**) was isolated as follows: one of the honey samples coming from orange trees (Sample K) (1 kg) was diluted with water (8 l) and the solution was passed through a column containing XAD4 resin (500 g) with a rate 10 ml/min. Then the resin was washed with MeOH (2 l) and the resulting solution was evaporated to a volume of 50 ml. Acetone (20 ml) was added, the solution was filtered and the filtrate was evaporated. The solid residue (2 g) was submitted to flash chromatography (CH_2Cl_2/MeOH gradient) to give (+)-hydroxylinalool (**2**) (5 mg) (Knapp et al., 1998).

2.3.4. (+)-Hydroxylinalool (2)

Colourless amorphous, $[\alpha]_D = +10°$ (MeOH, *c* 0.2). ^{1}H NMR ($CDCl_3$): δ 1.30 (3H, s, CH_3-6), 1.59 (2H, m, H-5), 1.65 (3H, d, $J = 1$ Hz, CH_3-2), 2.08 (2H, m, H-4), 3.98 (2H, s, H-1), 5.07 (1H, dd, $J = 11.0$, 1.5 Hz, H-8a), 5.22 (1H, dd, $J = 17.5$, 1.5 Hz, H-8b), 5.41 (1H, tq, $J = 7.0$, 1.0, H-3), 5.91 (1H, dd, $J = 17.5$, 11.0 Hz, H-7). ^{13}C NMR ($CDCl_3$): δ 68.7 (C-1), 135.0 (C-2), 125.8 (C-3), 22.3 (C-4), 41.7 (C-5), 73.3 (C-6), 144.8 (C-7), 111.8 (C-8), 13.6 (CH_3-2), 27.8 (CH_3-6). EI-MS: $m/z = 152$ $(M-46)^+$, 137, 119, 93, 79, 71 (100%).

The (R/S) 3-hydroxy-4-phenyl-2-butanone (**1**) was isolated as follows: one of the honey samples coming from thyme (sample

A) (1 kg) was diluted with water (1 l) and the solution was extracted with petroleum ether (1 l). The petroleum ether extract (50 mg) was submitted to flash chromatography, using cyclohexane/diethylether (from 90:10 to 50/50 gradient) to give (R/S) 3-hydroxy-4-phenyl-2-butanone (8 mg) (Awano et al., 1995).

*2.3.5. (R/S) 3-Hydroxy-4-phenyl-2-butanone (**1**)*

Colourless amorphous, $[\alpha]_D = 0°$ ($CHCl_3$, *c* 0.2). ^1H NMR ($CDCl_3$): δ: 2.06 (3H, s, CH_3), 4.41(1H, ddd, *J* = 7.2, 5.2, 4.7 Hz, H-3), 2.85 (1H, dd, *J* = 7.2, 14.0 Hz, H-4a), 3.13 (1H, dd, *J* = 4.7, 14.0 Hz, H-4b), 3.38 (1H, d, *J* = 5.2 Hz, OH-3), 7.10 (5H, s, Ar-H). ^{13}C NMR ($CDCl_3$): δ 25.6 (C-1), 209.2 (C-2), 77.5 (C-3), 39.7 (C-4), 136.6 (C-5), 129.2 (C-6), 128.4 (C-7), 126.7 (C-8), 128.4 (C-9), 129.2 (C-10). EI-MS: *m*/*z* = 164 $(M)^+$, 146, 121, 103, 92, 91 (100%), 72, 65, 43.

Both compounds were identified by comparison of their $[\alpha]_D$, NMR and MS data with the literature values. All the fractions of the columns were estimated by TLC, using eluent systems similar to those used in the relative column chromatographies.

2.4. Antimicrobial assay

The isolated compounds **1** and **2**, as well as the combined extracts of the honey samples, were studied for their antimicrobial activity. *In vitro* antimicrobial studies were carried out by the dilution method, as previously described (Melliou & Chinou, 2005), measuring the MIC values in 96-hole plates against two Gram-positive bacteria: *Staphylococcus aureus* (ATCC 25923), *Staphylococcus epidermidis* (ATCC 12228), four Gram-negative bacteria: *Escherichia coli* (ATCC 25922), *Enterobacter cloacae* (ATCC 13047), *Klebsiella pneumoniae* (ATCC 13883) and *Pseudomonas aeruginosa* (ATCC 227853), as well as against three human pathogen fungi *Candida albicans* (ATCC 10231), *C. tropicalis* (ATCC 13801) and *C. glabrata* (ATCC 28838). In addition, the tests were performed against the oral pathogens (Gram-positive bacteria) *Streptococcus mutans* and *S. viridans*, obtained from the culture collection of the Laboratory of Microbiology of the Anticancer Hospital of Athens "St. Savvas". Stock solutions of the tested extracts and pure compounds were prepared at 10 and 1 mg/ml, respectively. Serial dilutions of the stock solutions in broth medium (100 µl of Müller-Hinton broth, or on Sabouraud broth for the fungi), were prepared in a microtitre plate (96 wells). Then 1 µl of the microbial suspension (the inoculum, in sterile distilled water) was added to each well. For each strain, the growth conditions and the sterility of the medium were checked and the plates were incubated as referred to above. MICs were determined as the lowest concentrations preventing visible growth. Standard antibiotic netilmicin solutions (at concentrations 4–88 µg/ml) were used in order to control the sensitivity of the tested bacteria. 5-Flucytocine and itraconazole solutions (at concentrations 0.5–25 µg/ml) were used as controls against the tested fungi (Sanofi, Diagnostics Pasteur at concentrations of 30, 15 and 10 µg/ml) whilst sanguinarine, an alkaloid, was used as a positive control for oral pathogens. For each experiment, any pure solvent used was also applied as a blind control. The experiments were repeated three times and the results were expressed as average values.

3. Results and discussion

3.1. 3-Hydroxy-4-phenyl-2-butanone

In five (samples A–E), amongst the examined Greek samples, 3-hydroxy-4-phenyl-2-butanone (**1**) was detected by GC–MS analysis in the petroleum ether extract of the honey samples (area percentages ranging from 13.0% to 68.3%). This compound was found only in thyme honey samples but not in all of them. In order to unambiguously identify the structure of **1**, the compound was then isolated and studied by spectral means (1D, 2D NMR). The isolated compound was optically inactive. Indeed, GC–MS study using a β-Dex se chiral stationary phase revealed that it was a 1:1 mixture of the 3R and 3S enantiomers (Fig. 2).

The above mentioned hydroxyketone **1** that has a characteristic "sweet" smell has been found as a natural product amongst the flower volatiles of *Wisteria floribunda* and *Mimusops elengi*, plants growing in Japan and Malaysia, respectively (Wong & Teng, 1994). The same compound has been found as a constituent of Australian *Eucalyptus* and leatherwood honeys (D'Arcy, Rintoul, Rowland & Blackman, 1997; Graddon, Morrison, & Smith, 1979; Rowland, Blackman, D'Arcy & Rintoul, 1995) and very recently in *Mentha* spp. honey (Jerkovic, Hegic, Marijanovic, & Bubalo, 2010) and in Sulla (*Hedysarum coronarium* L.) honey (Jerkovic, Tuberso, Gugic, & Bubalo, 2010). Our team had previously reported the isolation of compound **1** from Greek honeys (Melliou, Graikou, & Chinou., 2003) and, to our knowledge it is the first time that the isolation from honeys, structure determination and enantiomeric ratio of **1** is described.

Although compound **1** had been previously identified in honey types other than thyme honeys, it has been proposed as a possible chemical marker of thyme honey (Alissandrakis, Tarantilis, Harizanis, & Polissiou, 2007a; Aliferis, Tarantilis, Harizanis, & Alissandrakis, 2010; Alissandrakis, Tarantilis, Pappas, Harizanis, & Polissiou, 2009). According to these authors, its presence was attributed to the contribution of thyme nectar (Alissandrakis et al., 2009). In our case, compound **1** was identified only in unifloral thyme honey amongst the Greek honey samples, but not in all of them and, additionally, its origin could not be directly related to thyme flowers, as repeated GC–MS studies of thyme flower extracts revealed the total absence of **1**.

As 3-hydroxy-4-phenyl-2-butanone has a very characteristic smell, we examined its possible floral source. During investigations, amongst Greek plants with similar odours, found in the same ecosystem as thyme, we identified *Spartium junceum* (Fabaceae) as a possible source. Indeed, compound **1**, in the same enantiomeric ratio as in the thyme honeys, was identified in *Spartium junceum* flower extract. The chemical analysis of *Spartium junceum* essential oil has previously been reported (Miraldi, Ferri, & Giorgi 2004) but **1** was not identified as one of its constituents. In our hands too, **1** could not be detected in the essential oil of the flowers obtained by hydrodistillation and only as a minor constituent in the petroleum ether extract.

Although it is necessary to identify more possible floral sources of 3-hydroxy-4-phenyl-2-butanone, it is obvious that this compound is restricted to specific types of bee honey (especially thyme honey) but cannot be used as an exclusive chemical marker as has recently been proposed for thyme honeys (Alissandrakis et al., 2007a; Aliferis et al., 2010; Alissandrakis et al., 2009). The appearance of 3-hydroxy-4-phenyl-2-butanone is probably associated with other plants found in the same ecosystem, together with thyme, and flourishing at the same period of the year. It is very interesting that the same compound, but in different enantiomeric ratios, was also identified in two honey samples from Poland coming from completely different floral sources (*Taraxacum officinale*, *Trifolium repens*) (Skalicka, Melliou, Glowniak, & Chinou, 2004).

Additionally, the only compound that was identified in all thyme honey samples (A–F) was 3,4,5-trimethoxybenzaldehyde (area percentages ranging from 1.5% to 8.29%), confirming previous reports on thyme honey samples (Mannas & Altug, 2007). Other compounds detected in most, but not all, thyme samples were: phenylethanol, *p*-methoxyphenylethanol and 3,4,5-trimethoxybenzylmethyl ether.

3.2. 8-Hydroxylinalool

8-Hydroxylinalool (**2**) was detected by GC–MS chromatography only in three of the studied Greek honey samples coming from

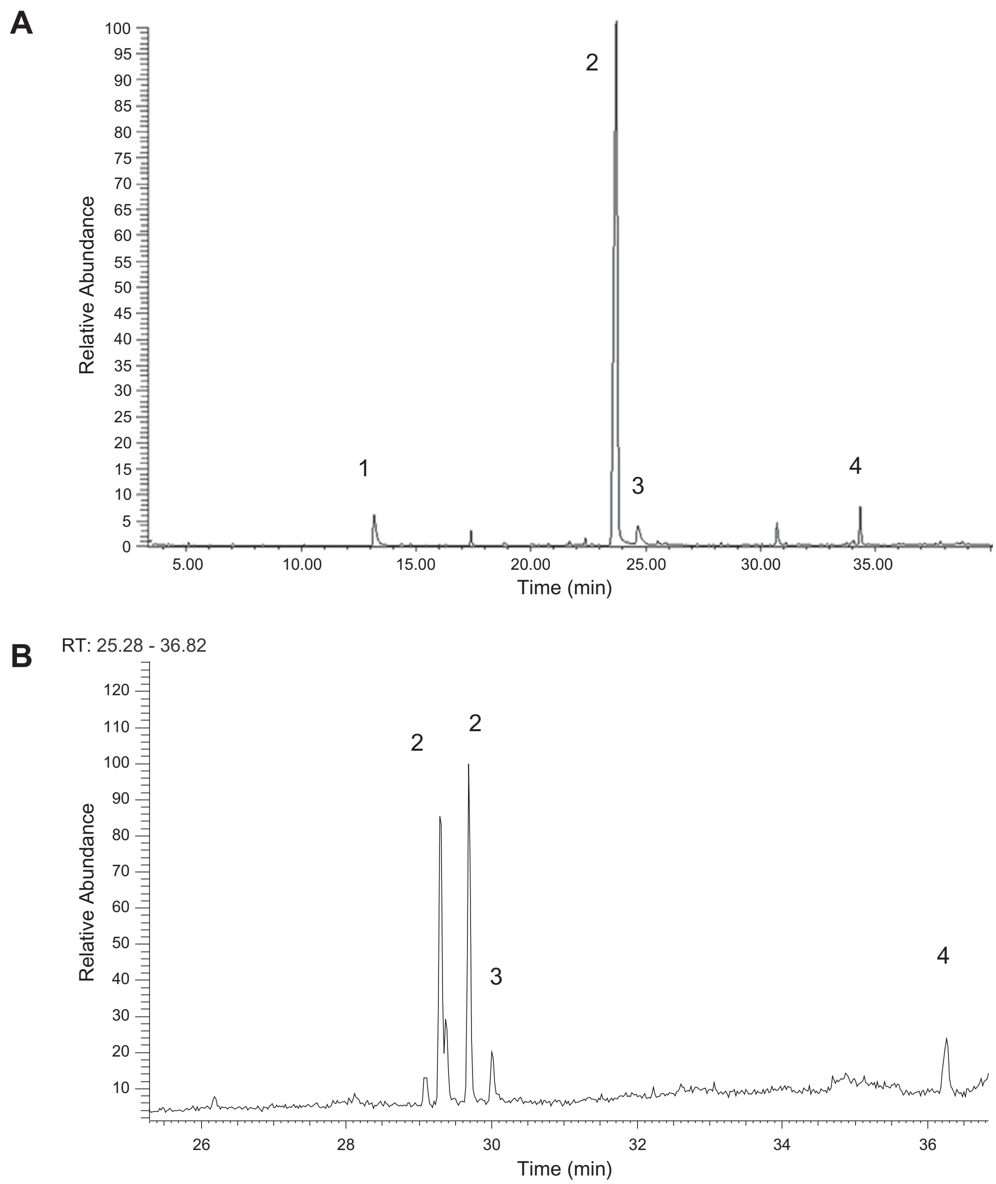

Fig. 2. GC–MS analysis of petroleum ether extract of honey sample A using non chiral (A) or chiral (B) column. (1) Phenylethanol, (2) (+/−)-3-hydroxy-4-phenyl-2-butanone, (3) p-methoxyphenylethyl alcohol, (4) 3,4,5-trimethoxybenzaldehyde.

orange trees. This compound was isolated and identified by NMR techniques as 8-hydroxylinalool. The optical rotation of the isolated compound was positive and GC–MS study using a β-Dex chiral stationary phase showed that it contained only one isomer.

Compound **2** has previously been identified in Nodding thistle honey from New Zealand (*Carduus nutans*) (Wilkins, Lu, & Tan, 1993) and very recently as a minor constituent in *Salix* spp. honey (Jerkovic & Marijanovic, 2010) and in *Acer* spp. honey (Jerkovic, Marijanovic, Malenica-Staver, & Lusic, 2010). This compound has also been identified in Spanish *Citrus* honey (Escriche, Visquert, Juan-Borras, & Fito, 2009) and in Greek *Citrus* honey (Alissandrakis, Daferera, Tarantilis, Polissiou, & Harizanis, 2003; Alissandrakis, Tarantilis, Harizanis, & Polissiou, 2007b) but without any data about its optical activity. In contrast to compound **1**, the origin of **2** could be directly associated with expected floral origin, the orange flowers. Orange flower extract, not only contained **2**, but, additionally, it showed enantiomeric purity identical to the corresponding compound **2** in honey extracts (Fig. 3). It should also be noted that, through the comparative study of the extracts of orange flowers and *Citrus* honey (samples G-L), (+)-8-hydroxylinalool, methyl anthranilate and caffeine, were determined in all studied cases as common constituents. Interestingly, although the predominant constituent of orange flowers extract was linalool (enantiomeric ratio −/+: 1/9 by GC–MS study, using a β-Dex chiral stationary phase), it was absent from all the studied *Citrus* honeys. Also, it should be noted that, although methyl anthranilate is a well known chemical marker of *Citrus* honey (Ferreres, Giner, & Tomas-Barberan, 1994; White & Bryant, 1996), its concentration has been reported to be quite variable (Papotti, Bertelli, Lolli, Sabatini, & Plessi, 2009) and moreover it has been identified in small quantities in honeys other than *Citrus* honey (Verzera, Campisi, Zappala, & Bonaccorsi, 2001). On the other hand, caffeine is considered as a characteristic of *Citrus* spp. flowers and it is noteworthy that it is restricted to only a few plant genera. Its presence in *Citrus* flowers and honeys (without discrimination between orange honey and other *Citrus* honey) has been previously reported (Trova, Cossa, & Gandolfo, 1994; Vacca, Agabbio, & Fenu, 1997).

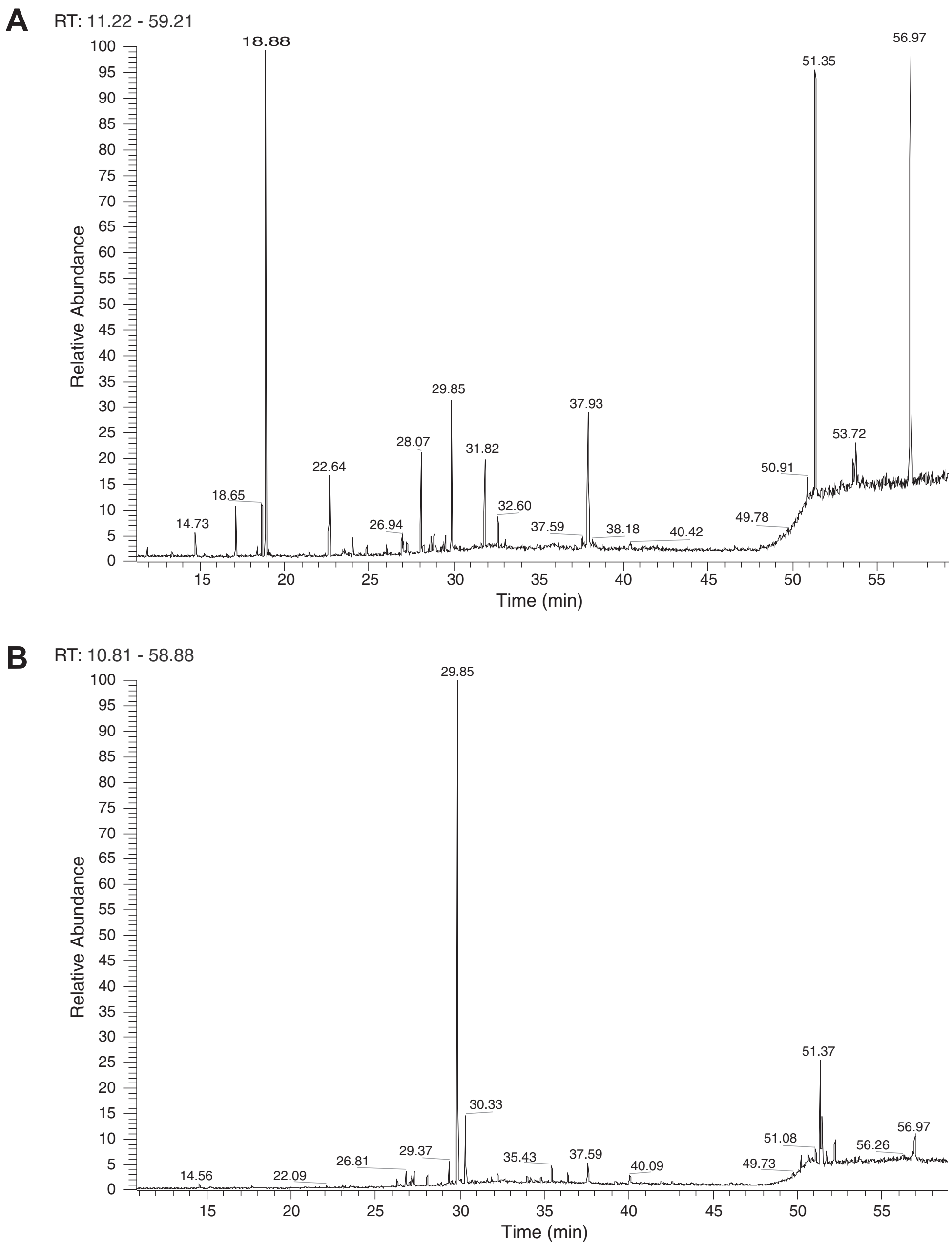

Fig. 3. GC–MS analysis of citrus flowers (A) and citrus honey sample K(B), using chiral column. 14.73: limonene, 18.65 (−)-linalool, 18.88: (+)-linalool, 28.07: methyl anthranilate, 29.85: (+)-8-hydroxylinalool, 30.33: furfural, 31.82: nerolidol, 37.93: *trans*-farnesol, 51.37: caffeine.

Based on the results of the 15 studied samples, it seems that the combination of (+)-8-hydroxylinalool with caffeine and methyl anthranilate could be used as a more reliable descriptive marker for the presence of *Citrus* honey.

The application of this combination of markers was very interesting in the case of two fir honey samples (samples N, K). In both cases, the presence of (+)-8-hydroxylinalool, caffeine and methyl anthranilate confirmed that the samples were adulterated with *Citrus* honey although the *Citrus* pollen could not be identified by pollen analysis.

3.3. Antimicrobial activity

All studied honey samples (unprocessed and unboiled), as well as the isolated compounds, were tested for their antimicrobial activity against human pathogenic Gram +/− bacteria and fungi, as well as oral pathogenic bacteria (Table 2). All samples showed very interesting antimicrobial profiles against the six standard Gram(+/−) bacteria strains and against the three pathogenic fungi, confirming the traditional reputation of honey as an antimicrobial agent. Especially, the thyme honey samples and the pure isolated compounds exerted powerful inhibitory effects. It is noteworthy, that *Gossypium* honey appeared as the least active, whilst *Pinus*, *Abies* and *Citrus* honey samples exhibited moderate activities against all assayed microorganisms. Through the antimicrobial screening, the characteristic compound **1**, proved to be active against all tested microorganisms (MIC values 0.35–3.2 mg/ml) whilst 8-hydroxy-linalool (**2**) exhibited the highest antimicrobial activity against all assayed microorganisms (MIC values 0.05–0.75 mg/ml), as well as against the oral pathogens *S. mutans* and *S. viridans*.

Table 2
Antimicrobial activities (mg/ml) of the studied honey extracts and pure compounds.

Studied samples	*S. aureus*	*S. epidermidis*	*P. aeruginosa*	*E. cloacae*	*K. pneumoniae*	*E. coli*	*S. mutans*	*S. viridans*	*C. albicans*	*C. tropicalis*	*C. glabrata*
Honey A	0.89	0.91	0.98	1.01	0.98	0.95	–	–	1.10	0.97	0.95
Honey B	0.65	0.67	0.75	0.94	0.87	0.74	–	–	1.18	1.00	0.90
Honey C	0.90	0.85	0.80	0.83	0.76	0.92	–	–	1.45	0.87	0.80
Honey D	0.56	0.60	0.45	1.10	0.45	0.72	–	–	0.79	0.75	0.67
Honey E	0.35	0.39	0.43	0.92	0.60	0.55	–	–	0.72	0.69	0.45
Honey F	0.44	0.43	0.49	0.95	0.53	0.49	–	–	0.65	0.58	0.35
Honey G	1.30	1.45	1.77	1.10	1.90	1.88	–	–	2.35	1.90	1.86
Honey H	1.20	1.25	0.89	1.30	0.98	0.49	–	–	1.65	1.58	1.35
Honey I	1.10	1.25	0.98	1.00	0.90	1.15	–	–	1.50	1.45	1.30
Honey J	1.76	1.80	1.50	1.75	1.70	1.67	–	–	1.88	1.80	1.77
Honey K	1.70	1.65	1.45	1.80	1.56	1.49	–	–	1.70	1.65	1.50
Honey L	1.68	1.82	1.52	1.77	1.60	1.55	–	–	1.76	1.67	1.46
Honey M	2.40	1.79	2.46	2.70	1.90	1.92	–	–	2.90	2.60	2.85
Honey N	1.25	1.18	0.94	1.24	0.75	0.89	–	–	1.88	2.25	1.80
Honey O	1.00	1.15	0.85	0.94	0.92	0.79	–	–	2.76	2.54	1.98
3-Hydroxy-4-phenyl-2-butanone (**1**)	0.35	0.72	3.10	1.25	2.84	2.15	–	–	3.20	2.85	2.33
8-Hydroxy-linalool (**2**)	0.08	0.05	0.25	0.35	0.50	0.75	0.09	0.05	0.75	0.50	0.35
Sanguinarine	–	–	–	–	–	–	0.015	0.015	–	–	–
Intraconazole	–	–	–	–	–	–	–	–	0.001	0.0001	0.001
5-Flucytocine	–	–	–	–	–	–	–	–	0.0001	0.001	0.01
Amphotericin B	–	–	–	–	–	–	–	–	0.001	0.0005	0.0004
Netilmicin	0.004	0.004	0.0088	0.008	0.008	0.01	–	–	–	–	–
AMCA	0.003	0.003	0.0031	0.0042	0.0048	0.005	–	–	–	–	–

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